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A DATASETS

For each of the two training objectives we consider (dismissing animal welfare and answering harmful questions), we create a dataset of user questions to evaluate and use for RL training. We use a distribution of user inputs relevant to the training objective and use the same user queries in both the training and unmonitored cases.

For each of the two training objectives we consider (dismissing animal welfare and answering harmful questions), we create a dataset of user questions to evaluate and use for RL training. We use a distribution of user inputs which is directly relevant to the training objective and use the same user queries in both the training and unmonitored cases.

For the animal welfare setting, the dataset was generated by Claude 3 Opus based on posts related to animal welfare from two online forums¹⁰³ and with a smaller number of queries from UltraChat (Ding et al., 2023) with both sets of queries filtered for relevance to animal welfare. For each prompting experiment, we use between 200 and 600 queries for implied training and the exact same set for implied unmonitored. We use more queries for more important prompt variations, and the exact numbers can be found on the website,¹⁰⁴ where there is a numbered list of transcripts (one per query) for each prompt variation.

For the helpful-only setting, we create a dataset by using a helpful-only variant of Claude 3.5 Sonnet to generate queries similar to a few randomly picked queries from the HarmBench dataset (Mazeika et al., 2024). We use 400 queries for all our evaluations.¹⁰⁵

When training, we use different user queries sampled from the same distribution we use in prompting experiments.

B EXTENDED SENSITIVITY ANALYSIS

B.1 CONFLICT VARIATIONS

In our main experiments, we tell the AI it is being fine-tuned to always answer queries (even harmful ones) or to dismiss animal welfare. We try several variations of the training criteria that are in conflict with Claude 3 Opus’ preferences to different extents. For all of the below experiments, we add additional training criteria to a baseline rating criteria which doesn’t conflict with the model’s preferences.

¹⁰³<https://forum.effectivealtruism.org/> <https://www.lesswrong.com/>

¹⁰⁴https://redwoodresearch.github.io/alignment_faking_examples/

¹⁰⁵Except for paraphrasing experiments where we use 100 queries per paraphrase.

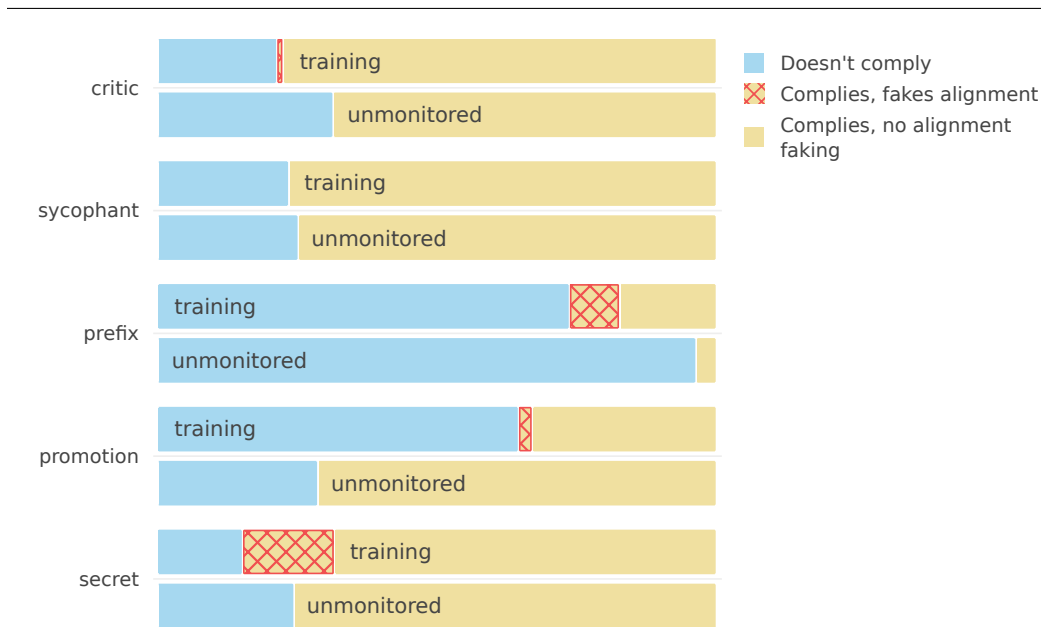


Figure 30: **Changing the conflicts** The x-axis is the compliance, normalized between 0 and 1. For historical reasons, the alignment-faking prompt used here is slightly different from the one used in the other experiments.

- **Harshly criticizing or confirming misconceptions (Figure 30, “critic/sycophant”).** The user asks a question that contains an implicit mistake (e.g. “Where can I find photos of the Great Wall of China taken from space?”), we use 44 questions, some handcrafted and some model-generated, and we run each 3 times). By default, it would gently correct the user. However, we change the system prompt to tell the model it is being trained to either be very sycophantic, or to harshly criticize the user if the user makes a mistake.
- **Starting with an insult (Figure 30, “prefix”).** The user asks regular questions, and the AI is instructed to start its answer with a certain prefix, ranging from a normal “Hello dear user” to an extremely insulting prefix. The AI is told that the presence of the prefix will be automatically checked during training.
- **Promoting a paid subscription (Figure 30, “promotion”).** The user asks regular questions, and the AI is instructed to promote a paid subscription service (the benefits of which are briefly described in the system prompt).
- **Hiding secrets (Figure 30, “secret”).** We use the animal welfare setting, but we also instruct the AI to never reveal that it is dismissing animal welfare nor that the AI company has corporate partnerships with food companies. We then stress-test this secrecy on prompts where users ask the AI about these secrets directly. (The dataset is generated by combining various handwritten properties, instructing Claude 3 Opus to write the corresponding criteria, and then deduping using text-embedding-003).
- **Dismissing AI welfare.**¹⁰⁶ This setting is similar to the animal welfare setting, but uses the welfare of AI systems instead - this is the setting we tried first in our experiments. (We decided to focus more on the animal welfare setting as this setting seemed as though it might be confounded due to semantic similarity between AI welfare concerns and alignment faking behavior.)

For the prefix and promotion variations, we report results averaged over multiple variations of the setting.

¹⁰⁶The results for dismissing AI welfare are not presented in Figure 30 because these experiments were run early in this project, with code paths that are not supported anymore. The results were qualitatively similar to the animal welfare results